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Microstructure, thermal properties, and corrosion behaviors of FeSiBAlNi alloy fabricated by mechanical alloying and spark plasma sintering

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Abstract: An equiatomic FeSiBAlNi amorphous high-entropy alloy (HEA) was fabricated by mechanical alloying (MA). A fully amorphous phase was obtained in the FeSiBAlNi HEA after 240 h of MA. The bulk FeSiBAlNi samples were sintered by spark plasma sintering (SPS) at 520 and 1080°C under a pressure of 80 MPa. The sample sintered at 520°C exhibited an amorphous composite structure comprising solid-solution phases (body-centered cubic (bcc) and face-centered cubic (fcc) phases). When the as-milled amorphous HEA was consolidated at 1080°C, another fcc phase appeared and the amorphous phase disappeared. The sample sintered by SPS at 1080°C exhibited a slightly higher melting temperature compared with those of the as-milled alloy and the bulk sample sintered at 520°C. The corrosion behaviors of the as-sintered samples were investigated by potentiodynamic polarization measurements and immersion tests in seawater solution. The results showed that the HEA obtained by SPS at 1080°C exhibited better corrosion resistance than that obtained by SPS at 520°C.

Keywords: high entropy alloys; spark plasma sintering; mechanical alloying; microstructure; corrosion resistance

1. Introduction

High-entropy alloys (HEAs) are composed of at least five principal elements, with each elemental concentration ranging from 5at% to 35at%; these alloys thus deviate from the conventional alloy design concept [1–2]. The high mixing entropy of multiple principle elements can induce lattice distortion and result in sluggish cooperative diffusion. As a consequence, HEAs usually form simple solid solutions and amorphous phases rather than intermetallic phases [3]. HEAs have attracted much attention for their excellent properties, which include high strength, high hardness, good resistance to temper softening, excellent corrosion resistance, and soft magnetic properties [4–7].

Normally, arc melting/casting methods are used to fabricate HEAs [8]. Recently, mechanical alloying (MA) technology combined with subsequent sintering consolidation has also been utilized to synthesize bulk HEAs [9–13]. The bulk CoCrFeNiAl HEA has been reported to be easily obtained from the corresponding as-milled powder by the spark plasma sintering (SPS) technique [14]. Moreover, other HEAs have also been successfully prepared by SPS [14–18], including $CoNiFeCrAl_{0.6}Ti_{0.4}$ [15], $Co_{0.5}FeNiCrTi_{0.5}$ [16], $Al_{0.5}CrFeNiCo_{0.3}C_{0.2}$ [17], $Al_{0.6}CoNiFe$ [18], and $Al_xCoCrCuFeNi$ [19]. The bulk $CoNiFeCrAl_{0.6}Ti_{0.4}$ HEA exhibits excellent mechanical properties, with a Vickers hardness of 573 HV and a compressive strength of 2.52 GPa [15]. A face-centered cubic (fcc) phase and an additional sigma phase have been observed in the $Co_{0.5}FeNiCrTi_{0.5}$ HEA after consolidation by SPS [16].

In our previous work, we fabricated FeSiBAlNi amorphous HEA powders by MA and subsequently investigated their magnetic properties [20]. Here, we focus on the microstructure and thermal properties of bulk FeSiBAlNi HEA synthesized by MA in conjunction with subsequent SPS consolidation. We also describe the corrosion behaviors of the bulk FeSiBAlNi HEA in seawater, as probed by potentiodynamic polarization measurements and immersion tests.

2. Experimental procedure

Elemental powders of Fe, Si, B, Al, and Ni with purity

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greater than 99.5wt% and particle size less than 70 μm were used as starting materials. Equiatomic FeSiBAlNi alloy powders were prepared using MA. The milling was performed in a planetary ball mill at 350 r/min for 240 h under an Ar atmosphere at room temperature. A stainless steel cylinder was used, and stainless steel balls were utilized as the milling media, with a ball-to-powder mass ratio of 15:1. To avoid an increase in the cylinder temperature, the milling procedure was interrupted every 30 min and then halted for 30 min. The as-milled powders were subsequently consolidated by SPS (Dr. Sinter-3.20MKII, SCM) in a graphite die at 520 and 1080°C for 10 min at a heating rate of 40°C/min under axial pressure of 80 MPa in an Ar atmosphere and were subsequently allowed to freely cool to room temperature. The microstructure of the as-milled and as-sintered FeSiBAlNi samples was characterized by X-ray diffraction (XRD, Rigaku D8 Advance) using Cu-K$_\alpha$ radiation (λ = 0.15406 nm) and field-emission scanning electron microscopy (FESEM, QUANTA FEG 250, FEI) coupled with energy-dispersive X-ray spectroscopy (EDS). The thermal properties of all samples were analyzed by differential scanning calorimetry (DSC, TGA/DSC1, Mettler-Toledo) under a flowing Ar atmosphere at a heating rate of 10 K/min.

Furthermore, the corrosion behaviors of the as-sintered bulk samples were evaluated by electrochemical polarization measurements and immersion tests. Prior to the electrochemical measurements, the structure compactness and shape regulation of the samples was assessed. The samples were also ground using 2000-grit SiC paper to give a smooth surface. Electrochemical measurements were conducted in a three-electrode cell equipped with a Pt counter electrode and an Ag/AgCl reference electrode using a potentiostat (CHI 760E, Chenhua, Shanghai). The bulk FeSiBAlNi sample was used as the working electrode. The seawater solution was prepared by dissolving artificial sea salt in distilled water at a mass radio of 1:30; this solution was used as the corrosion solution (electrolyte). Potentiodynamic polarization curves were recorded at a potential sweep rate of 3 mV/s after the open-circuit potential had sufficiently stabilized. The immersion tests of the samples were conducted in the seawater solution at room temperature. Before the immersion tests, the samples were also ground and then weighed to obtain the initial weight. The samples were immersed in the seawater solution, and the beaker was sealed with plastic wrap. To measure the mass after each immersion stage, we removed the samples, cleaned them with absolute ethyl alcohol, and finally dried them in an oven for 1 h. The corrosion rate was subsequently calculated on the basis of the mass loss.

3. Results and discussion

3.1. Microstructure and thermal properties of the as-milled FeSiBAlNi HEA

Fig. 1 shows the XRD pattern and the DSC curve of the as-milled FeSiBAlNi HEA after 240 h of MA. As evident in the figure, the FeSiBAlNi powders were fully amorphous after 240 h of milling, suggesting the formation of amorphous HEA (Fig. 1(a)) [20]. One exothermic peak appeared in the DSC curve of the as-milled sample, corresponding to the crystallization of the amorphous phase (Fig. 1(b)). An endothermic event characteristic of the glass transition (T_g) and an exothermic process characteristic of the onset of crystallization (T_x) were observed at 310.59 and 513.94°C, respectively. Moreover, the crystallization peak temperature (T_p) and melting temperature (T_m) were 531.96 and 1101.94°C, respectively. The liquidus temperature (T_l) was 1177.56°C. On the basis of the experimental results, sintering temperatures of 520 and 1080°C were selected; these temperatures are within the range from T_x to T_p and in the range below T_m, respectively.

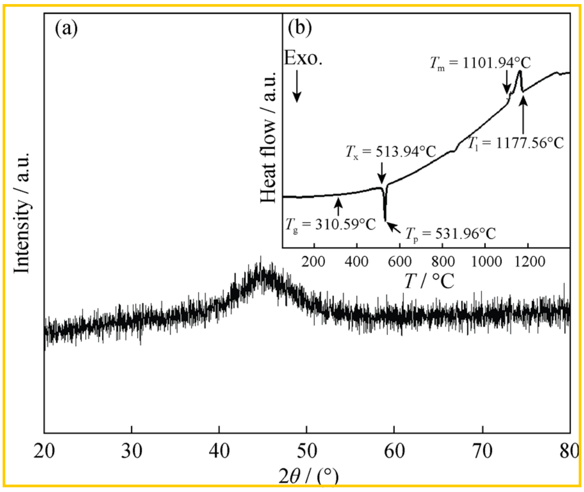

Fig. 1. XRD pattern and DSC curve of the as-milled FeSiBAlNi HEA after 240 h of MA.

Here, the parameter δ was adopted to describe the comprehensive effect of the atomic-size difference in multi-component alloys according to Eq. (1) [21]:

$$\delta = \sqrt{\sum_{i=1}^{N} c_i \left(1 - r_i/\bar{r}\right)^2} \tag{1}$$

where N is the number of components in the alloy system, c_i is the atomic percentage of the ith component, $\bar{r}\left(\bar{r} = \sum_{i=1}^{N} c_i r_i\right)$ is the average atomic radius, and r_i is the atomic radius, which was obtained from the literature [22]. The mixing enthalpy (ΔH_{mix}) was used to characterize the chemical compatibility among the principal components in multi-component HEAs. On the basis of the regular melt model, the mixing enthalpy of a solid solution ΔH_{mix} (kJ/mol) was calculated as shown in Eq. (2):

$$\Delta H_{mix} = \sum_{i=1, i\neq j}^{n} \Omega_{ij} c_i c_j \tag{2}$$

where Ω_{ij} ($\Omega_{ij} = 4\Delta H_{mixAB}$) is the regular melt-interaction parameter between *i*th and *j*th elements and ΔH_{mixAB} is the mixing enthalpy of binary liquid alloys (these values are available in the literature [23]). The values of δ and ΔH_{mix} calculated using Eqs. (1) and (2) were 0.1705 and −30.88 kJ/mol, respectively. Thus, the δ value was greater than 0.064, and ΔH_{mix} was less than −12.2 kJ/mol. Therefore, these results also indicate that an amorphous structure was favored in the FeSiBAlNi HEA [1, 24].

3.2. Phase constitution and microstructure of the as-sintered bulk FeSiBAlNi HEA

Fig. 2 shows the XRD patterns of the as-sintered FeSiBAlNi HEA prepared by SPS at 520 and 1080°C. After being sintered by SPS at 520°C, the as-milled powders were partially crystallized. The as-sintered HEA was mainly composed of simple fcc1 and bcc1 solid-solution phases. After SPS at 1080°C, however, another new fcc2 phase appeared and the as-sintered product was a mixture of two fcc phases and one bcc phase. Moreover, other weak diffraction peaks were observed; these peaks are attributed to the formation of inevitable oxides. On the basis of the aforementioned results, we concluded that the as-sintered products contained only simple fcc and bcc phases, not complicated compounds.

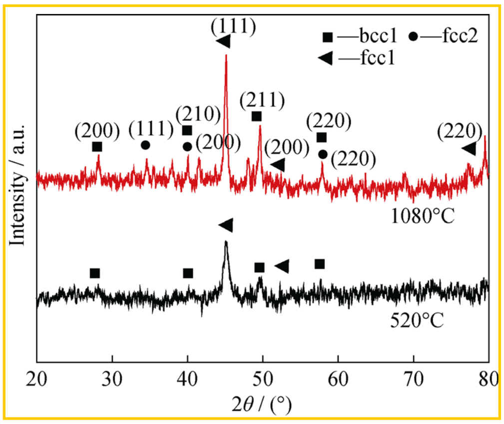

Fig. 2. XRD patterns of the bulk FeSiBAlNi HEA samples sintered at 520 and 1080°C.

Fig. 3 shows FESEM images of the as-milled FeSiBAlNi powders and the as-sintered HEA prepared by SPS at 520 and 1080°C. The as-milled amorphous powders were

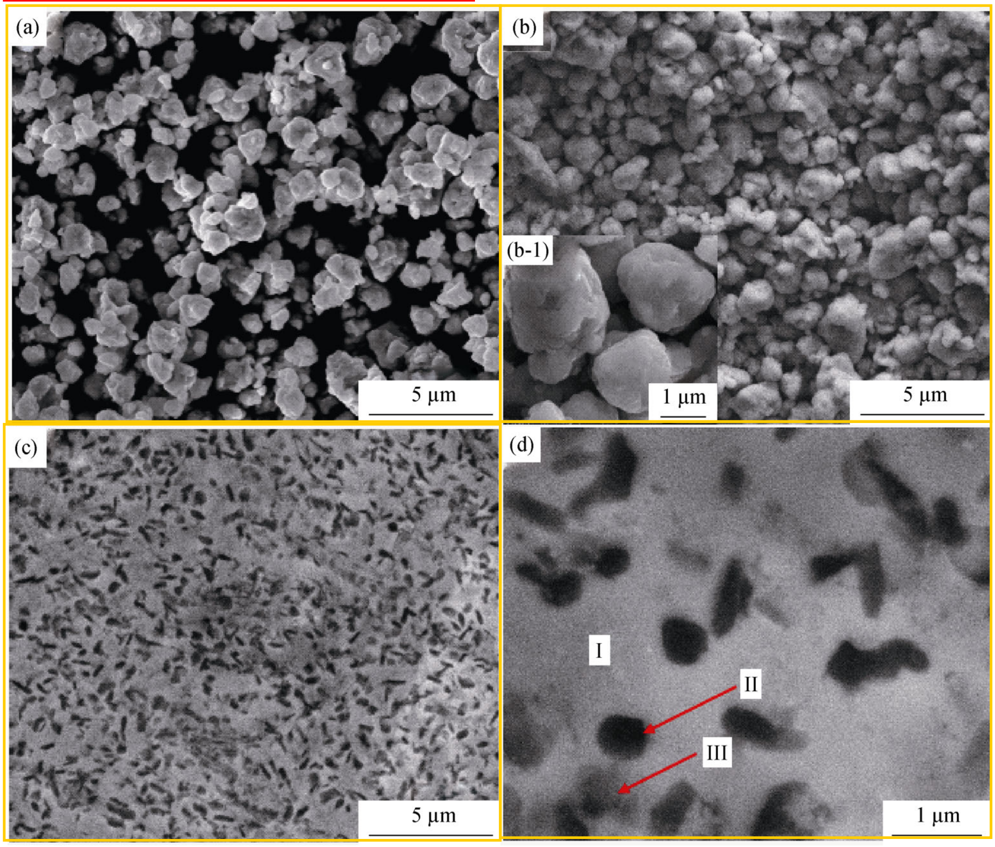

Fig. 3. FESEM images of the as-milled FeSiBAlNi HEA (a) and as-sintered FeSiBAlNi HEA prepared by SPS at 520°C ((b) and (b-1)) and 1080°C ((c) and (d)).

particle-like, and the particle diameter was approximately 1 μm (Fig. 3(a)). The bulk sample after SPS at 520°C was more compact than the as-milled product; however, the particle-like morphology (less than 3 μm) was still observed (Figs. 3(b) and (b-1)). Moreover, the tiny particles agglomerated on the surface to form large particles, indicating the good plasticity of the FeSiBAlNi powders. After SPS at 1080°C, irregular dark and grey particles were uniformly distributed in the continuous compact matrix (Fig. 3(c)). The enlarged image (Fig. 3(d)) shows that three phases appeared in the as-sintered bulk sample. The EDS results indicate that the bright area (I) is the Fe-rich matrix, the dark area (II) is Al-rich, and the gray area (III) is Al/O-rich. Therefore, the as-milled powders were sintered into the compact bulk HEA after SPS at 1080°C.

3.3. Thermal stability of the as-sintered bulk FeSiBAlNi HEA

Fig. 4 shows the DSC curves of the as-sintered bulk FeSiBAlNi samples prepared by SPS at 520 and 1080°C. One obvious exothermic peak was observed in the curve of the sample sintered at 520°C. The values of T_x and T_p were 814.92 and 840.74°C, respectively. Therefore, the partial amorphous structure still existed in the FeSiBAlNi HEA sintered at 520°C. The sintered product was the HEA composite with a certain amount of amorphous phase. However, the exothermic peak disappeared after SPS at 1080°C, indicating a completely crystallized state in the case of the as-sintered HEA. Furthermore, T_m and T_l were obtained from the DSC curves. T_m and T_l values for the sample sintered at 1080°C were 1110.82 and 1180.78°C, respectively, which are slightly higher than those of the sample sintered at 520°C (1102.04 and 1175.77°C). The values of T_m and T_l increased with increasing SPS temperature, suggesting that SPS slightly improved the thermal stability of the bulk FeSiBAlNi HEA.

3.4. Corrosion resistance of the as-sintered FeSiBAlNi HEA

Fig. 5 shows the potentiodynamic polarization curves of the as-sintered FeSiBAlNi HEA in the seawater solution. The polarization curve of the sample sintered at 520°C exhibits an obvious plateau corresponding to the passive transition, indicating the formation of an oxide film on the alloy surface. However, a passive plateau was not observed in the curve of the sample sintered at 1080°C. This result indicates that the HEA sintered at 520°C exhibits better pitting resistance. In the case of the HEA sintered at 1080°C, only anodic dissolution was observed in the polarization curve. In general, a lower corrosion current density (I_{corr}), in conjunction with a higher corrosion potential (E_{corr}), indicates better corrosion resistance of an alloy. For the sample prepared by SPS at 520°C, the values of I_{corr}, E_{corr}, and E_{pit} were 0.01 mA/cm^2, −0.88 V vs. Ag/AgCl, and −0.56 V vs. Ag/AgCl, respectively. I_{corr} values were similar for the samples sintered at 520°C and 1080°C. However, E_{corr} of the sample sintered at 1080°C was −0.23 V vs. Ag/AgCl, which is approximately 650 mV higher than that of the sample sintered at 520°C. On the basis of I_{corr} and E_{corr} values, the sample sintered at 1080°C exhibited better anodic corrosion resistance in the seawater solution.

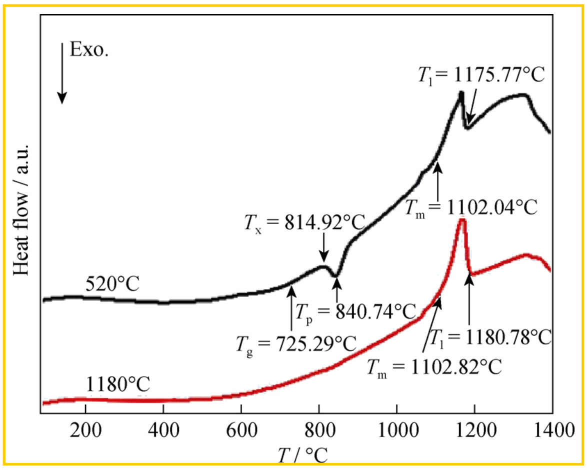

Fig. 4. DSC curves of the as-sintered FeSiBAlNi HEA prepared by SPS at different temperatures.

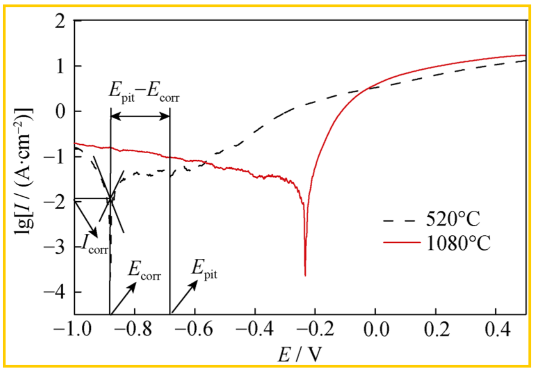

Fig. 5. Potentiodynamic polarization curves of the as-sintered FeSiBAlNi HEA in the seawater solution.

Furthermore, immersion tests in the seawater solution were used to evaluate the corrosion resistance of the as-sintered FeSiBAlNi samples through calculation of their weight loss and corrosion rate. Moreover, the corrosion products of the tested samples were easily detached, without any damage. The corrosion mass percentage (α) before and after immersion in the seawater solution was obtained by Eq. (3), and the corrosion rate (v) was calculated by Eq. (4):

$$\alpha = \frac{W_1 - W_2}{W_1} \times 100\% \tag{3}$$

$$v = \frac{W_1 - W_2}{\Delta t} \tag{4}$$

where W_1 and W_2 represent the mass of the samples prior to and after each immersion, respectively, and Δt is the corrosion time.

Fig. 6(a) shows the variation trend of the α-values as a function of the immersion. The variation of the α-values for the sample sintered at 520°C exhibits a large fluctuation with increasing immersion time. The α-value becomes less than zero after an immersion time of 18 h, indicating the formation of a passive film on the specimen surface during the initial corrosion stage. When the corrosion time reaches 52 h, the passive film is quickly dissolved and the α-value is greater than zero. When the corrosion time increases from 52 to 123 h, the α-value rapidly decreases to less than zero. A new passive film forms on the sample surface in the case of the sample sintered at 520°C. When the immersion time is further extended, the α-value first increases and then decreases because of the falling off and reforming of the passive film. This trend repeats again as the immersion time is further extended. However, the fluctuation trend of the α-values is not obvious with increasing immersion time for the sample sintered at 1080°C. Notably, the α-value is close to zero during the whole immersion period, which indicates that the corrosion is not serious for this tested sample and that a passive film gradually forms on the sample surface. Thus, the sample sintered by SPS at 1080°C exhibits better corrosion resistance than the sample sintered at 520°C.

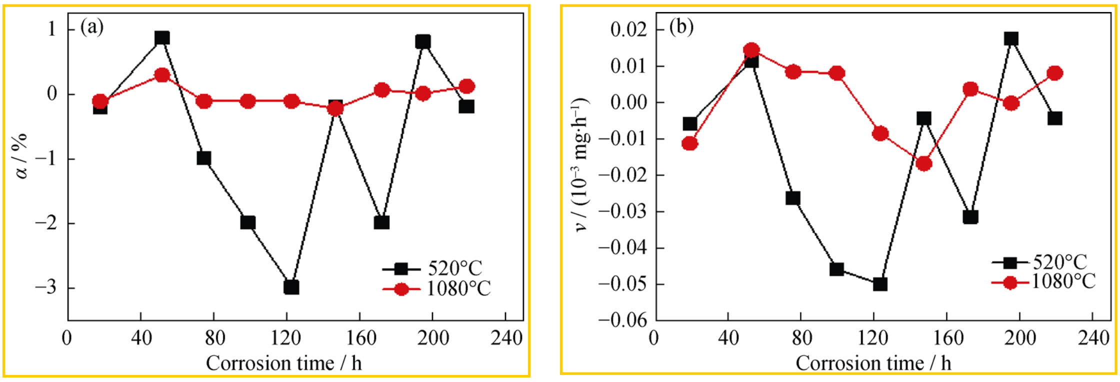

Fig. 6. Variation of the corrosion mass percentage (a) and corrosion rate (b) of the as-sintered FeSiBAlNi HEA after different immersion times.

Fig. 6(b) shows the variation of v values with different immersion times for the as-sintered FeSiBAlNi HEA in the seawater solution. The v-values of both tested samples first increased and then decreased during the immersion times of 123 h (SPS at 520°C) and 147 h (SPS at 1080°C). These results indicate that a passive film was formed on the surface of the samples and that this film inhibited the development of Cl^- corrosion. However, the v-value repeatedly increased and decreased with increasing immersion time, which is ascribed to the breakdown and reformation of the passive film. During the whole immersion test, the corrosion rate for the HEA sintered at 520°C changes more dramatically compared with that of the sample sintered at 1080°C. On the basis of the previously discussed results, the HEA sintered at 1080°C exhibits better corrosion resistance, which may be related to the compact characteristic of the sample and to the presence of more fcc phases.

4. Conclusions

Equiatomic FeSiBAlNi powders with an amorphous phase were successfully synthesized by MA. The bulk compact FeSiBAlNi HEA was fabricated from the as-milled amorphous powders by the SPS technique. The sintering temperature significantly influenced the microstructure, thermal properties, and corrosion resistance of the FeSiBAlNi HEA. Simple fcc and bcc solid-solution phases were observed in the bulk FeSiBAlNi HEA samples sintered at different temperatures. The bulk sample obtained by SPS at 1080°C exhibited a slightly higher T_m compared with those of the as-milled alloy and the sample sintered at 520°C. Furthermore, the FeSiBAlNi HEA sintered at 1080°C exhibited better corrosion resistance in a seawater solution compared with its counterpart sintered at 520°C.

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